

OC Core Methods and Reproducibility Companion v1.3.3

Ontology of Continua Core release review artifact

Alexander Yashin

Version 1.3.3

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1 OC Core Methods and Reproducibility Companion v1.3.3

1.1 Publication Identity

Author. Alexander Yashin, Independent Researcher, ORCID 0009-0008-6166-0914. **Research instrument.** Logion is the research-instrument and institute-automation system used to prepare, check, package, and audit the work; it is not an author. **Methodological framework.** ESTRA is the methodological framework used in the work; it is not an author or affiliation. **Version.** 1.3.3. **Release identity.** oc_core_1_3_3. **Concept DOI.** 10.5281/zenodo.17899134. **Artifact role.** Methods, validation, reproducibility, and replay companion. **Intended audience.** readers checking reproducibility, formalization, finite semantics, validation, and artifact traceability.

1.2 Dedication

Dedicated to my dear wife Maria, without whom this work would have been impossible.

1.3 Acknowledgements

Substantive review and idea acknowledgements. They are acknowledged for review pressure, ideas, criticism, or external response that improved the work. Acknowledgement does not imply authorship, endorsement, publication approval, or agreement with the theory’s release form.

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- Eduard Fadeev
- Gennady Alekseevich Nosov
- Sergey Shpadyrev
- Stanislav Tsukrov

1.4 Abstract

This companion explains how the release is checked: source traceability, proof and finite-model artifacts, validation, simulation, counterexample search, replay, checksums, and reproducibility limits.

1.5 Reader Contract

Read this document when checking whether a claim can be replayed, traced, or bounded by negative controls and reproducibility limits. The release promotes evidence-bound model-core claims and excludes unsupported complete-science closure, unrestricted all-domain numerical completion, or unrestricted superiority-over-modern-science claims.

1.6 Table of Contents

The generated PDF contents provide page locations. The reader-facing route is:

1. Publication identity
2. Dedication
3. Acknowledgements
4. Abstract
5. Reader contract
6. Reading map
7. Block 5: Define Claim Classes
8. Block 5: Define Formal Proof Standard
9. Block 5: Define Executable Semantics Standard
10. Block 5: Define Empirical Evidence Standard
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37. Block 17: Define Checksums
38. Block 17: Define Replay Protocols
39. Block 17: Define Source-to-PDF Trace
40. Block 17: Define Reproducibility Limits
41. Additional sections continue in document order.

2 Body

2.1 Block 5

2.1.1 Define Claim Classes

Define Claim Classes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Claim Classes. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Claim Classes to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Claim Classes states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background

research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Claim Classes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.1.2 Define Formal Proof Standard

Define Formal Proof Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Formal Proof Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

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2.1.3 Define Executable Semantics Standard

Define Executable Semantics Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Executable Semantics Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed

to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

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Synthesize Executable Semantics Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.1.4 Define Empirical Evidence Standard

Define Empirical Evidence Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Empirical Evidence Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Empirical Evidence Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

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2.1.5 Define Simulation

Define Simulation and Counterexample Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Simulation and Counterexample Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Simulation and Counterexample Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

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Synthesize Simulation and Counterexample Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.1.6 Define Negative Controls

Define Negative Controls and Falsifiers is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the

definition_model route for Negative Controls and Falsifiers. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

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2.1.7 Define Source, Artifact,

Define Source, Artifact, and Traceability Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Source, Artifact, and Traceability Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Source, Artifact, and Traceability Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

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2.1.8 Define Editorial

Define Editorial and Reviewer Evidence Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Editorial and Reviewer Evidence Standard. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

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2.2 Block 10

2.2.1 Define Formalization Strategy

Define Formalization Strategy is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Formalization Strategy. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

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2.2.2 Define Lean/Lake Subset

Define Lean/Lake Subset is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Lean/Lake Subset. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

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2.2.3 Define Typed Structures in Lean

Define Typed Structures in Lean is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Typed Structures in Lean. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

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established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.4 Define Machine-Checked Theorem Subset

Define Machine-Checked Theorem Subset is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Machine-Checked Theorem Subset. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Machine-Checked Theorem Subset to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Machine-Checked Theorem Subset states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Machine-Checked Theorem Subset synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.5 Define Finite Model Semantics

Define Finite Model Semantics is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Finite Model Semantics. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Finite Model Semantics to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical

computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Finite Model Semantics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Finite Model Semantics synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.6 Define Witness Cases

Define Witness Cases is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Witness Cases. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Witness Cases to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Witness Cases states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Witness Cases synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on

reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.7 Define Tamper

Define Tamper and Negative Controls is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Tamper and Negative Controls. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Tamper and Negative Controls to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Tamper and Negative Controls states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Tamper and Negative Controls synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.2.8 Define Limits of the Formalized Subset

Define Limits of the Formalized Subset is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Limits of the Formalized Subset. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Limits of the Formalized Subset to proof, evidence, or replay carries the evidential burden for

the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Limits of the Formalized Subset states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Limits of the Formalized Subset synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3 Block 11

2.3.1 Define Evidence Ledger

Define Evidence Ledger is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Evidence Ledger. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Evidence Ledger to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Evidence Ledger states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence

without inflating it.

Synthesize Evidence Ledger synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.2 Define Mathematics Anchor

Define Mathematics Anchor is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Mathematics Anchor. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Mathematics Anchor to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Mathematics Anchor states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Mathematics Anchor synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.3 Define Physics Evidence Lane

Define Physics Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Physics Evidence Lane. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims

without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Physics Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Physics Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Physics Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.4 Define Chemistry Evidence Lane

Define Chemistry Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Chemistry Evidence Lane. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Chemistry Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Chemistry Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The

governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Chemistry Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.5 Define Biology Evidence Lane

Define Biology Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Biology Evidence Lane. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Biology Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Biology Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Biology Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.6 Define Systems

Define Systems and Civilizational Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Systems and Civilizational Evidence Lane. The paragraph draws on

formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Systems and Civilizational Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Systems and Civilizational Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Systems and Civilizational Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.7 Define Simulations

Define Simulations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Simulations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Simulations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Simulations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to over-

state scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Simulations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.8 Define Adversarial Simulations

Define Adversarial Simulations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Adversarial Simulations. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Adversarial Simulations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Adversarial Simulations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Adversarial Simulations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.9 Define Target-Blind

Define Target-Blind and Held-Out Evidence is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Target-Blind and Held-Out Evidence. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Target-Blind and Held-Out Evidence to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Target-Blind and Held-Out Evidence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Target-Blind and Held-Out Evidence synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.3.10 Define Evidence Promotion Boundaries

Define Evidence Promotion Boundaries is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Evidence Promotion Boundaries. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Evidence Promotion Boundaries to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited

proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Evidence Promotion Boundaries states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Evidence Promotion Boundaries synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4 Block 17

2.4.1 Define Build Environment

Define Build Environment is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Build Environment. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Build Environment to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Build Environment states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Build Environment synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize

established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.2 Define Data Availability

Define Data Availability is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Data Availability. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Data Availability to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Data Availability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Data Availability synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.3 Define Code Availability

Define Code Availability is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Code Availability. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Code Availability to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computa-

tional evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Code Availability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Code Availability synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.4 Define Artifact Inventory

Define Artifact Inventory is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Artifact Inventory. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Artifact Inventory to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Artifact Inventory states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Artifact Inventory synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on

reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.5 Define Checksums

Define Checksums and Integrity is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Checksums and Integrity. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Checksums and Integrity to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Checksums and Integrity states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Checksums and Integrity synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.6 Define Replay Protocols

Define Replay Protocols is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Replay Protocols. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Replay Protocols to proof, evidence, or replay carries the evidential burden for the surround-

ing claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Replay Protocols states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Replay Protocols synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.7 Define Source-to-PDF Trace

Define Source-to-PDF Trace is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Source-to-PDF Trace. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Source-to-PDF Trace to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Source-to-PDF Trace states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Source-to-PDF Trace synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.4.8 Define Reproducibility Limits

Define Reproducibility Limits is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Reproducibility Limits. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Reproducibility Limits to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Reproducibility Limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Reproducibility Limits synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5 Block 18

2.5.1 Define Versioning

Define Versioning and Release Identity is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Versioning and Release Identity. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote

empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Versioning and Release Identity to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Versioning and Release Identity states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Versioning and Release Identity synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.2 Define Public Release Asset Set

Define Public Release Asset Set is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Public Release Asset Set. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Public Release Asset Set to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Public Release Asset Set states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The

governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Public Release Asset Set synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.3 Define Journal Package Map

Define Journal Package Map is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Journal Package Map. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Journal Package Map to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Journal Package Map states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Journal Package Map synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.4 Define Citation

Define Citation and Metadata Governance is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Citation and Metadata Governance. The paragraph draws on formal

model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Citation and Metadata Governance to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Citation and Metadata Governance states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Citation and Metadata Governance synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.5 Define Publication Boundary

Define Publication Boundary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Publication Boundary. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Publication Boundary to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Publication Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission

to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Publication Boundary synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.6 Define Owner Approval Boundary

Define Owner Approval Boundary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Owner Approval Boundary. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Owner Approval Boundary to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Owner Approval Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Owner Approval Boundary synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.7 Define Incident

Define Incident and Known-Error Lessons is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Incident and Known-Error Lessons. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Incident and Known-Error Lessons to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Incident and Known-Error Lessons states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Incident and Known-Error Lessons synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.5.8 Define External Use Guidance

Define External Use Guidance is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for External Use Guidance. The paragraph draws on formal model and foundation and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind External Use Guidance to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses empirical computational evidence, formal model and foundation, machine checked and finite semantics; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited

proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for External Use Guidance states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by empirical computational evidence, review attack response and limits. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize External Use Guidance synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

2.6 Block 999

2.6.1 Reproducibility Structure

Reproducibility Structure and Audience is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Reproducibility Structure and Audience. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.2 Axiom 0.9 (Finite local structural complexity)

Axiom 0.9 (Finite local structural complexity) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.3 (F9.18) Method collapse under paradigm shift

(F9.18) Method collapse under paradigm shift is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F9.18) Method collapse under paradigm shift. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.4 Lean Formalization Subset

Lean Formalization Subset carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.5 FM-T133-HYBRID-GUARD-NONBOOLEAN-NEG

FM-T133-HYBRID-GUARD-NONBOOLEAN-NEG carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.6 Formal Methods

Formal Methods and Lean carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.7 Lean formal subset

Lean formal subset carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.8 Lean Build

Lean Build carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.9 Witness method

Witness method carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.10 Executable Finite-Model Evidence

Executable Finite-Model Evidence carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.11 Finite semantic witnesses

Finite semantic witnesses carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.12 Simulation

Simulation and Counterexample Search carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.13 Empirical Validation Matrix

Empirical Validation Matrix and Replay Ledger carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.14 Meta-Criteria

Meta-Criteria and Cross-Domain Validation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.15 Unified validation pipeline

Unified validation pipeline carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.16 Type~V Experiments: Validation of $E(K_{-1})E(K_{-1})$

Type~V Experiments: Validation of $E(K_{-1})E(K_{-1})$ carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.17 Type~VI Experiments: Validation of the Operator $F_{-1} 2$

Type~VI Experiments: Validation of the Operator $F_{-1} 2$ carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.18 Type~VII Experiments: Validation of the Operator $F_{-2} 3$

Type~VII Experiments: Validation of the Operator $F_{-2} 3$ carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.19 Experimental validation

Experimental validation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.20 Experimental Validation Pipeline

Experimental Validation Pipeline carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.21 Method ladder

Method ladder carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.22 (P10.17) Meta-stability Requires Finite Cycles

(P10.17) Meta-stability Requires Finite Cycles carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.23 (P2.3) Prediction of Finite Spatial Neighbourhoods

(P2.3) Prediction of Finite Spatial Neighbourhoods carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.24 (P3.3) Finite-Range Causality

(P3.3) Finite-Range Causality carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.25 (P8.17) Institutional Reproduction Cycle

(P8.17) Institutional Reproduction Cycle carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.26 OC 1.3.1 Simulation

OC 1.3.1 Simulation and Data Appendix carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.27 Scientific Workflow Provenance

Scientific Workflow Provenance and Reproducibility Systems carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.28 Validation QA

Validation QA carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.29 Reviewer Response Method

Reviewer Response Method states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.30 Adversarial Review Method

Adversarial Review Method states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.31 Epistemic Governance, Methodology,

Epistemic Governance, Methodology, and Scientific Cycles synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.32 Axiom 0.9 (Finite local structural complexity)

Axiom 0.9 (Finite local structural complexity) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.33 Type~II Experiments: Hyper-Transfinite Modal Spaces $\hat{}(12)$

Type~II Experiments: Hyper-Transfinite Modal Spaces $\hat{}(12)$ carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.34 Falsifiability via Methodological Structure

Falsifiability via Methodological Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

2.6.35 (F9.16) Methodological incoherence

(F9.16) Methodological incoherence synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.36 (F9.18) Method collapse under paradigm shift

(F9.18) Method collapse under paradigm shift synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.37 (F9.19) Methodological overfitting

(F9.19) Methodological overfitting synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.38 (F9.20) Methodological-epistemic mismatch

(F9.20) Methodological-epistemic mismatch synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.39 Editorial Method for the Guide

Editorial Method for the Guide synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named.

The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.40 OC133-NUM-MATH-FINITE

OC133-NUM-MATH-FINITE carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.41 Reproducibility Method

Reproducibility Method synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.42 Simulation

Simulation and adversarial simulation carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.43 Finite Semantic Checks

Finite Semantic Checks carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.44 Minimum Reproducibility Interpretation

Minimum Reproducibility Interpretation synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route;

it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.45 Observed Reproducibility Bill of Materials

Observed Reproducibility Bill of Materials synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on historical toc recovery but does not add new scientific strength beyond the evidence already named. The boundary remains: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.46 Methods

Methods is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for Methods. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.47 Type~II Experiments: Hyper-Transfinite Modal Spaces $\hat{}$ (12)

Type~II Experiments: Hyper-Transfinite Modal Spaces $\hat{}$ (12) carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses historical toc recovery; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.48 Falsifiability via Methodological Structure

Falsifiability via Methodological Structure states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by historical toc recovery. The governing boundary is: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.49 (F9.16) Methodological incoherence

(F9.16) Methodological incoherence is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F9.16) Methodological incoherence. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.50 (F9.19) Methodological overfitting

(F9.19) Methodological overfitting is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F9.19) Methodological overfitting. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.6.51 (F9.20) Methodological-epistemic mismatch

(F9.20) Methodological-epistemic mismatch is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to recover the historical scientific content route for (F9.20) Methodological-epistemic mismatch. The paragraph draws on historical toc recovery and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: historical recovery restores structure and source route; it does not promote stronger claims without current evidence. The next paragraph continues the same scientific route while preserving source trace and claim boundary.

2.7 Source Trace

Exact source bindings, path hashes, quality scorer hooks, and transition records are recorded in the review package manifest. They are kept out of the main prose to avoid turning the document into a ledger dump.